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34 1 Abstract

35 **Background.** Inference about the biomarker-treatment interaction in aggregated
36 N-of-1 trials depends on the analyst’s test procedure: classical repeated-measures
37 ANOVA (RM-ANOVA), the linear mixed-effects model (LME), or generalized esti-
38 mating equations (GEE). A common but rarely examined feature of the classical
39 analysis is that it requires the continuous biomarker to be dichotomized. We
40 quantify the consequences of test-procedure choice for the interaction, separating
41 the cost of dichotomization from the cost of the test procedure proper.

42 **Methods.** We derive the closed-form strict RM-ANOVA F -statistic for the
43 biomarker-stratified treatment-by-time interaction in a balanced split-plot design
44 with categorical predictors and the sphericity assumption, and relate it to the
45 Wald t -test for the corresponding fixed-effects coefficient in a linear mixed model
46 with continuous-time AR(1) residual correlation. We then conduct a Monte Carlo
47 simulation study, designed and reported under the¹ ADEMP framework, that
48 crosses five test procedures (strict RM-ANOVA on the dichotomized biomarker;
49 a continuous-biomarker within-subject contrast; the standard pmsimstats linear-
50 mixed analysis with `corCAR1`; and GEE with naive and with Mancl-DeRouen
51 small-sample sandwich variance) against the biomarker-moderation coefficient
52 $c_{bm} \in \{0, 0.45\}$ and sample size $N \in \{25, 35, 70, 100\}$, a 40-cell factorial at the
53 prazosin-PTSD reference design with 1000 replicates per power cell and 2000 per
54 Type I cell. The continuous-biomarker contrast differs from strict RM-ANOVA
55 only in retaining the continuous biomarker, isolating the cost of dichotomization.
56 The cycle-by-period design grid is specified here but its empirical sweep is deferred
57 to a companion paper.

58 **Results.** The closed-form strict RM-ANOVA test agrees with the linear-mixed
59 Wald test under sphericity and balanced design. In the simulation, most of the
60 power gap between strict RM-ANOVA and the mixed model is the cost of di-
61 chotomization rather than of the test procedure: at $N = 25$, replacing the di-
62 chotomized biomarker with the continuous one raises power by 0.180, while the
63 further step to the full mixed model adds only 0.027, a pattern stable across
64 N (dichotomization 0.08–0.21, test procedure at most 0.04). Among the four
65 continuous-biomarker procedures, the linear-mixed analysis attains the highest
66 power within nominal Type I error; GEE with the naive sandwich inflates Type
67 I error to between $1.5\times$ and $1.9\times$ nominal across all four sample sizes, which the

68 Mancl-DeRouen correction removes at a moderate (about 10-percentage-point)
69 power cost while restoring 93–95% interval coverage. The inline Mancl-DeRouen
70 standard error agrees with the `geesmv` reference to within 2.5%.

71 **Conclusions.** The analyst’s most consequential decision is to keep the biomarker
72 continuous rather than dichotomize it; among continuous-biomarker procedures
73 the linear-mixed analysis is the preferred default and a dichotomization-free
74 within-subject contrast is a close, transparent alternative. Naive GEE should
75 not be used at the sample sizes examined without a small-sample correction.
76 This paper resolves the test-procedure axis at a fixed reference design; the joint
77 cycle-by-period design grid is specified here and deferred to a companion paper.

78 2 Introduction

79 2.1 A claim about how design and test choices interact

80 The present paper develops the methodological case for treating the analyst’s
81 two principal choices in an aggregated N-of-1 trial jointly rather than separately.
82 The two choices are the test procedure for the biomarker-treatment interaction
83 (strict repeated-measures ANOVA, the linear mixed-effects model with continuous-
84 time AR(1) residual correlation, the mixed-effects model for repeated measures,
85 generalized estimating equations) and the trial-design parameters that govern
86 within-subject treatment cycling (number of treatment cycles, on-drug and off-
87 drug period lengths, on/off symmetry, optional run-in and run-out phases). The
88 validity and power of any given test depend on what the design assumes about
89 carryover, autocorrelation, and within-subject variance structure; conversely, the
90 optimal design parameters depend on which test the analyst plans to use. Treating
91 the two as independent optimizations, as the published methodological literature
92 typically does, produces locally optimal but jointly suboptimal combinations, and
93 we shall argue that this gap is consequential at the sample sizes characteristic of
94 precision-medicine pilot studies.

95 2.2 Why the question matters for chronic-condition pharmacol- 96 ogy

97 In this section we shall set out three reasons why the joint treatment of test
98 procedure and design parameters is warranted specifically for the aggregated N-
99 of-1 design class.

100 We begin with the physiological motivation. Within-participant residuals in
101 chronic-condition trials are autocorrelated not as an analytic convenience but
102 because most chronic-condition response trajectories exhibit serial correlation on
103 a timescale set by drug pharmacokinetics, condition-specific natural-history drift,
104 and measurement reactivity (Hawthorne, structured-visit, and diary-completion
105 effects). Continuous-time AR(1) residual correlation is, in our view, the appro-
106 priate structural prior for this physiology, but it does not satisfy the sphericity
107 assumption required by strict RM-ANOVA.² show through simulation that un-

corrected RM-ANOVA inflates Type I error under sphericity violations; they also report, importantly, that the Greenhouse-Geisser-corrected RM-ANOVA and the multilevel linear model perform comparably once the correction is applied, and that the multilevel model is not uniformly safer (it can be the more liberal of the two in some conditions). The degrees-of-freedom corrections^{3,4} recover nominal levels only conditionally.⁵ extend this evidence with finer epsilon resolution, with the conventional rule favoring Greenhouse-Geisser when $\epsilon < 0.75$. The advantage we claim for the mixed-effects model with explicit `corCAR1` residual structure is thus not that it is uniformly more powerful than a well-corrected RM-ANOVA, but that it models the AR(1) structure directly, accommodates the continuous biomarker and unequal time spacing, and so avoids both the dichotomization and the degrees-of-freedom penalty that the classical route incurs.

2.3 What the published N-of-1 methodology literature does and does not do

We begin with the state of the analytic-model literature. The systematic review of⁶ documents that 83.8% of the published N-of-1 studies they catalogue ignore autocorrelation entirely;⁷ and⁸ reach similar conclusions in neurology and behavioral-medicine subdomains. CONSORT extension reporting standards^{9,10} codify reporting quality but do not address the analytic-model decision in any detail, and¹¹ identifies the analytic-method gap as the highest-priority methodological problem in the field. It appears, then, that the majority of N-of-1 analyses are conducted under a misspecified residual covariance assumption, and that the field has not yet converged on remedies.

Test-procedure comparisons for longitudinal data are not new. The generic contrast of RM-ANOVA against GEE and mixed-effects models is well established:¹², for instance, report through simulation that RM-ANOVA is appreciably less powerful than GEE and mixed models for continuous longitudinal endpoints, the same ordering we recover. What that literature does not do is treat the biomarker-treatment interaction estimand, the aggregated N-of-1 / cross-over design, or the continuous-time AR(1) structure together, nor does it isolate the dichotomization cost that the classical route imposes specifically when the moderator is continuous. The MMRM consensus established by Mallinckrodt and colleagues^{13–16} applies to parallel-group longitudinal trials with monotone missingness, not to aggregated N-of-1 designs. The GEE methodology of¹⁷ with sandwich variance estimation has been adapted to small-cluster settings via bias-corrected estimators^{18,19}, but the small-sample concerns are acute at trial-relevant N-of-1 cluster counts.²⁰ report that in stepped-wedge cluster trials with fewer than six independent units only 6.6% of analyses use appropriate small-sample corrections, illustrating the gap between method availability and practice. None of these papers address the biomarker-interaction estimand in the cross-over or N-of-1 setting directly.

The gap, then, is this. Across the recent systematic reviews^{6,7,21} and the methodological literature surveyed above, no published study crosses test procedure (strict RM-ANOVA, MMRM, GEE, mixed-effects with `corCAR1`) against a design-

parameter grid (cycles, period lengths, asymmetry, run-in, run-out) under simulation for the aggregated N-of-1 design class. The present paper endeavors to address that gap, at least on the test-procedure axis.

2.4 Run-in design and the placebo-selection controversy

2.5 What this paper contributes

We make four contributions, each addressed in a section below.

First, we derive the closed-form strict RM-ANOVA F -statistic for the biomarker-stratified treatment-by-time interaction in a balanced split-plot design with categorical predictors and the sphericity assumption (§2). We relate the strict-RM-ANOVA test to the Wald t -test on the biomarker-treatment fixed-effects coefficient in a linear mixed model with continuous-time AR(1) residual correlation, identifying the conditions on residual covariance under which the two tests agree and the regimes in which they diverge. The derivation extends standard cross-over methodology^{22,23} to the biomarker-stratified case and clarifies the connection to the contemporary sphericity-violation literature^{2,5}.

Third, we conduct a Monte Carlo simulation study (§4–§5) that crosses five test procedures (strict RM-ANOVA on the dichotomized biomarker; a continuous-biomarker within-subject contrast; the standard pmsimstats linear-mixed analysis with `corCAR1`; and GEE with naive and with Mancl-DeRouen²⁴ small-sample sandwich variance) against the biomarker-moderation coefficient and sample size at the fixed reference design. The continuous-biomarker contrast is the dichotomization-free analogue of the strict RM-ANOVA test and lets us separate the cost of dichotomizing the biomarker from the cost of the test procedure proper. The simulation runs on the pmsimstats package data-generating process under Architecture A (mean moderation) via `generateData()` and `lme_analysis()`, holding fixed the response-curve and covariance-structure axes addressed in `analysis/report/07-gompertz-evaluation/` and `analysis/report/01-dgp-mean-moderation-vs-mvn/`. Results are reported in the¹ ADEMP framework adopted as the project-wide simulation-reporting standard.

3 Closed-form strict RM-ANOVA test of the biomarker-treatment interaction

In this section we shall derive the closed-form strict RM-ANOVA F -statistic for the biomarker-treatment interaction in a balanced split-plot design with categorical predictors and the sphericity assumption, and relate it to the Wald t -statistic on the corresponding linear-mixed coefficient. The derivation specializes classical split-plot machinery²⁵ to the biomarker-stratified case in the simplest design that supports the test: two biomarker strata, two treatment phases, and T within-phase timepoints. The presentation is deliberately confined to strict RM-ANOVA. Modern extensions — the mixed-effects model for repeated measures²⁶, general-

191 ized estimating equations¹⁷, and ANCOVA with random subject effects — appear
 192 only for context; the closed-form expression we report is the textbook compound-
 193 symmetric form. Three findings anchor the rest of the paper:

- 194 1. Under sphericity and balance, the strict RM-ANOVA test of the biomarker-
 195 treatment interaction is identical to a two-sample t -test on the within-
 196 subject treatment differences, with degrees of freedom $2(n - 1)$.
- 197 2. The closed form yields a clean non-central F power formula that we use as
 198 the analytic anchor for the simulation study in §4-§5.
- 199 3. Five assumptions of the closed form are violated by the pmsimstats data-
 200 generating process (DGP; continuous biomarker, AR(1) within-subject co-
 201 variance, exponential carryover, missing data, unequal time spacing). Each
 202 violation favors the linear-mixed analysis over the strict RM-ANOVA test in
 203 operating characteristics, motivating the primary-and-sensitivity reporting
 204 strategy of §6.

205 3.1 Notation and design

206 Specifically, we consider the following structural elements:

- 207 • Two biomarker strata $i \in \{1, 2\}$, formed by dichotomizing the continuous
 208 biomarker at a pre-specified cut-point. Strict RM-ANOVA requires cat-
 209 egorical predictors; the dichotomization is the price of admission to the
 210 framework.
- 211 • n subjects per stratum, balanced. Index $l \in \{1, \dots, n\}$.
- 212 • Two treatment phases $j \in \{1, 2\}$, active drug and placebo. In an N-of-1
 213 design these are two phases through which every subject passes; in a parallel-
 214 group design they are between-subject and the design is not a split-plot.
- 215 • T within-phase timepoints, indexed $k \in \{1, \dots, T\}$. In the simplest case
 216 $T = 1$, the analyst pre-averages within-phase observations to a single mean
 217 per subject per phase.

218 We write Y_{ijlk} for the symptom outcome at biomarker stratum i , treatment phase
 219 j , subject l within stratum i , and timepoint k . The strict classical split-plot model
 220 is

$$\begin{aligned}
 Y_{ijlk} = & \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \pi_{l(i)} \\
 & + \tau_k + (\alpha\tau)_{ik} + (\beta\tau)_{jk} + (\alpha\beta\tau)_{ijk} + e_{ijlk},
 \end{aligned}
 \tag{1}$$

221 with sum-to-zero constraints on each fixed effect, the random subject effect $\pi_{l(i)} \sim$
 222 $N(0, \sigma_\pi^2)$ nested in stratum, and within-subject error $e_{ijlk} \sim N(0, \sigma_e^2)$. The
 223 model presupposes the classical sphericity (compound symmetry) condition on
 224 the within-subject covariance: every pair of within-subject observations has the
 225 same covariance σ_π^2 , and every observation has the same total variance $\sigma_\pi^2 + \sigma_e^2$.

226 3.2 Sums of squares and the interaction F -statistic

227 We begin the derivation by partitioning the total sum of squares. The total sum
228 of squares partitions into between-subjects and within-subjects components:

$$SS_{\text{tot}} = SS_{\text{between}} + SS_{\text{within}}. \quad (2)$$

229 The between-subjects component decomposes further into SS_A (biomarker stratum) and $SS_{S(A)}$ (subject within stratum):

$$SS_A = bnT \sum_{i=1}^a (\bar{Y}_{i..} - \bar{Y}_{...})^2, \quad SS_{S(A)} = bT \sum_{i=1}^a \sum_{l=1}^n (\bar{Y}_{i.l} - \bar{Y}_{i...})^2. \quad (3)$$

231 The within-subjects component decomposes into SS_B (treatment phase), SS_{AB}
232 (the biomarker-treatment interaction, the primary target of the test), and a within-
233 subjects error term $SS_{B \times S(A)}$:

$$SS_{AB} = nT \sum_{i,j} (\bar{Y}_{ij..} - \bar{Y}_{i...} - \bar{Y}_{.j..} + \bar{Y}_{...})^2, \quad (4)$$

$$SS_{B \times S(A)} = T \sum_{i,j,l} (\bar{Y}_{ijl.} - \bar{Y}_{i.l.} - \bar{Y}_{ij..} + \bar{Y}_{i...})^2. \quad (5)$$

234 Corresponding degrees of freedom are $df_A = a - 1$, $df_{S(A)} = a(n - 1)$, $df_B =$
235 $b - 1$, $df_{AB} = (a - 1)(b - 1)$, and $df_{B \times S(A)} = a(n - 1)(b - 1)$, the last being
236 the within-subjects error degrees of freedom appropriate for the treatment effect
237 and its interaction with the between-subjects factor. The biomarker-treatment
238 interaction is then tested by the F -ratio

$$F_{AB} = \frac{MS_{AB}}{MS_{B \times S(A)}} = \frac{SS_{AB}/df_{AB}}{SS_{B \times S(A)}/df_{B \times S(A)}}. \quad (6)$$

239 Under the null $H_0 : (\alpha\beta)_{ij} \equiv 0$ and the sphericity assumption, $F_{AB} \sim$
240 $F(df_{AB}, df_{B \times S(A)})$.

241 3.2.1 The 2×2 split-plot case

242 We now specialize to $a = b = 2$ (two biomarker strata, two phases), setting $T = 1$
243 for cleanest exposition; that is, the analyst pre-averages within-phase observations
244 to a single value per subject per phase before computing the interaction. The
245 sample interaction contrast is

$$\hat{L} = (\bar{Y}_{11.} - \bar{Y}_{12.}) - (\bar{Y}_{21.} - \bar{Y}_{22.}), \quad (7)$$

246 That is, $\hat{L}$ is the difference of within-subject treatment differences across
247 biomarker strata, and is the natural biomarker-treatment interaction statistic.

248 The interaction parameter under the sum-to-zero ANOVA parameterization
 249 satisfies $(\widehat{\alpha\beta})_{11} = \hat{L}/4$ (and the other three cells take values $\pm\hat{L}/4$), so the
 250 interaction sum of squares is $SS_{AB} = nT\hat{L}^2/4$, and with $df_{AB} = 1$ this equals
 251 the interaction mean square. Under sphericity and balance, $MS_{B \times S(A)} = \hat{\sigma}_e^2$.
 252 Substituting into equation (6) yields

$$F_{AB} = \frac{nT\hat{L}^2}{4\hat{\sigma}_e^2}, \quad df_1 = 1, \quad df_2 = 2(n-1). \quad (8)$$

253 3.2.2 Equivalence to a within-subject t -test

254 We note that equation (8) is equivalent to a two-sample t -test on the within-subject
 255 differences. Define $\Delta_{il} = \bar{Y}_{i1l} - \bar{Y}_{i2l}$ as the within-subject treatment difference for
 256 subject l in biomarker stratum i , averaged over T within-phase timepoints. Then

$$\hat{L} = \bar{\Delta}_1 - \bar{\Delta}_2, \quad \bar{\Delta}_i = \frac{1}{n} \sum_{l=1}^n \Delta_{il}. \quad (9)$$

257 Under sphericity, Δ_{il} are i.i.d. within stratum with $\text{Var}(\Delta_{il}) = 2\sigma_e^2/T$, since the
 258 random subject effect cancels in the within-subject difference. The variance of $\hat{L}$
 259 is therefore

$$\text{Var}(\hat{L}) = \frac{2 \cdot 2\sigma_e^2}{nT} = \frac{4\sigma_e^2}{nT}, \quad (10)$$

260 and the t -statistic for the contrast is

$$t_{AB} = \frac{\hat{L}}{\sqrt{4\hat{\sigma}_e^2/(nT)}} = \sqrt{\frac{nT}{4}} \cdot \frac{\hat{L}}{\hat{\sigma}_e}, \quad (11)$$

261 3.3 Relation to the linear-mixed Wald test

262 We shall now relate the strict-RM-ANOVA F -statistic of equation (8) to the pm-
 263 simstats `nlme::lme` Wald t -statistic on the `bm:Dcb` coefficient. The two are asymp-
 264 totically equivalent under three conditions: (i) the biomarker is dichotomized
 265 at the cut-point that defines the strata, (ii) the within-subject covariance is ex-
 266 actly compound-symmetric, and (iii) `Dcb` is binary, the A1 specification of the
 267 carryover-sensitivity manuscript rather than the matched-decay A2 specification.
 268 When all three conditions hold, the two tests reach the same coefficient and the
 269 same standard error up to small-sample correction factors. When any one fails,
 270 the mixed-model test extracts more information than the strict RM-ANOVA: the
 271 continuous biomarker contributes within-stratum variation, the AR(1) covariance
 272 is modeled directly, and the continuous exposure-decay predictor draws on the
 273 off-drug trajectory.

274 The pmsimstats DGP imposes AR(1) within-factor correlation, which is not
 275 compound-symmetric. Under AR(1), Box's ϵ is strictly below 1, and treating

the full multi-timepoint series as repeated within-subject levels under classical RM-ANOVA inflates Type I error unless a degrees-of-freedom correction is applied. The relevant magnitude is recorded in the project’s own audit (`docs/06-ar1-residual-correlation.tex`): at $T = 8$ timepoints, fitting a mixed model with no residual-correlation structure (the naive analogue of treating AR(1) data as exchangeable) produces Type I error of 13 to 17 percent for the crossover (CO) and open-label (OL) designs, which switching to `nlme::lme` with `corCAR1` returns to nominal. We stress that this 13-to-17-percent figure is for the unstructured/naive fit, not for the strict RM-ANOVA arm evaluated in this paper: that arm pre-averages each phase to a single value (the 2×2 form of §2.2), so its within-subject covariance is 2×2 , sphericity holds by construction, and its Type I error is at nominal (§5). The sphericity penalty is therefore a property of the un-pre-averaged classical analysis; for the pre-averaged strict RM-ANOVA the operative cost is dichotomization, not sphericity. Either way the linear-mixed analysis with `corCAR1` avoids both costs, modeling the AR(1) structure directly rather than absorbing it via a degrees-of-freedom penalty.

3.3.1 Power calculation under the closed form

For the 2×2 case, equation (8) yields a closed-form power formula that we shall use as the analytic anchor for the simulation study in §4–§5. Under the alternative $H_1 : \hat{L} = \delta$, the statistic $F_{AB} = nT \hat{L}^2 / (4\sigma_e^2)$ follows a non-central F distribution with non-centrality parameter

$$\lambda = \frac{nT \delta^2}{4\sigma_e^2}, \quad (12)$$

and power at level α is

$$1 - \beta = \Pr \left[F_{AB} > F_{1,2(n-1),1-\alpha} \mid \lambda \right], \quad (13)$$

3.3.2 Limitations of the strict classical formulation

Unfortunately, the closed-form simplicity of equation (8) is bought at the price of five strong assumptions, each of which is empirically problematic in the pmsimstats setting. We present them in approximate order of severity.

1. *Categorical biomarker.* The pmsimstats biomarker (resting SBP in the prazosin-PTSD application) is continuous and has no defensible cut-point. Dichotomizing at the median or at a clinical threshold loses the within-stratum biomarker variation, attenuates the interaction estimator, and inflates Type II error. In the carryover-sensitivity production data the median split gives a $\hat{L}$ standard error roughly 1.4 to 1.7 times that of the continuous-biomarker linear-mixed coefficient, translating to a 30 to 50 percent power penalty.
2. *Sphericity.* Compound-symmetric within-subject covariance is empirically

rejected for time-series symptom data: nearby timepoints are more correlated than distant ones. The AR(1) structure pmsimstats implements explicitly violates sphericity. After Greenhouse-Geisser correction, the test recovers nominal Type I error but loses degrees of freedom and power.

3. *No carryover model.* Strict RM-ANOVA has no provision for residual drug effect during the off-drug period; the model treats off-drug observations as if no drug had been administered. Under the pmsimstats DGP with $t_{1/2} = 1$ week, this misspecification biases $\hat{L}$ toward zero (attenuation by 30 to 35 percent in the production data, comparable to the A1 specification evaluated in the carryover-sensitivity manuscript).

4. *Balanced design required.* Equations (6) and (8) assume equal stratum sizes and equal numbers of within-phase timepoints. Real trials have missing data; classical RM-ANOVA handles missingness only through complete-case deletion. The MMRM framework’s likelihood-based handling of missing-at-random (MAR) data²⁶ is genuinely better-behaved.

5. *Equal time spacing.* Strict RM-ANOVA treats time as a categorical factor with implicitly equal-spaced levels. Real N-of-1 trials have unequally-spaced visits; pmsimstats implements continuous-time AR(1) (`corCAR1`), which the strict-RM-ANOVA framework cannot accommodate.

These limitations motivate retaining the linear-mixed analysis with continuous-time AR(1) residual correlation as primary, and reporting the strict-RM-ANOVA closed-form F -statistic as a sensitivity comparator. The five violations are not equally consequential: the categorical-biomarker and no-carryover assumptions are the most problematic in the pmsimstats setting, together accounting for a 30 to 50 percent attenuation of the interaction estimator relative to the linear-mixed analysis. The recommendation operationalized in §6 follows this strategy.

4 Cycle-by-period design grid

The cycle-by-period design grid that varies the number of treatment cycles, on-drug and off-drug period lengths, and on/off symmetry around the²⁷ hybrid reference is specified in `docs/27-cycle-period-design-sweep-plan.md` and constitutes the second axis of the joint design-by-test sensitivity question framed in §1. The empirical sweep across this grid is deferred to a follow-up paper. We shall note here, in the interest of transparency, that the present paper narrows its scope to the test-procedure axis at the prazosin/PTSD reference design (Hendrickson hybrid open-label plus blinded discontinuation, OL+BDC; $N \in \{25, 35, 70, 100\}$, $c_{bm} \in \{0, 0.45\}$), which the simulation infrastructure does support at production rep counts. The §6 recommendation is correspondingly scoped to test-procedure choice rather than the joint (cycle count, period length, test procedure) optimum that the design-by-test sensitivity question requires. The follow-up paper executing the cycle-by-period sweep is the natural extension and is recorded as a portfolio item.

5 Simulation design

The simulation program follows the ADEMP framework of¹. The primary estimand is the biomarker-treatment interaction inference under each of the five test procedures. Four operate on the continuous biomarker and share the same interaction estimand $\beta_{bm:D}$ on the symptom scale: a within-subject treatment-effect contrast regressed on the continuous biomarker (the dichotomization-free analogue of the two-sample t -test of equation (11)); the linear-mixed-effects Wald t on $\hat{\beta}_{bm:D}$; the GEE Wald z on $\hat{\beta}_{bm:D}$ with naive sandwich variance; and the GEE Wald z with Mancl-DeRouen small-sample sandwich variance per²⁴. The fifth, the strict classical RM-ANOVA F -statistic, operates on the median-dichotomized biomarker and so targets a dichotomized estimand on a different scale. The continuous-biomarker contrast is included specifically to separate the cost of dichotomization (strict RM-ANOVA versus the contrast, both pre-averaged to one value per phase) from the cost of the test procedure proper (the contrast versus the longitudinal linear-mixed and GEE analyses). Primary inferential outputs are power $\pi = \Pr(p < 0.05)$ for $H_0 : \beta_{bm:D} = 0$ at $\alpha = 0.05$, and Type I error under $\beta_{bm:D} = 0$. Secondary estimands, reported for the four coefficient-bearing arms, are bias of $\hat{\beta}_{bm:D}$ relative to a high-precision Monte-Carlo reference value of the estimand, the empirical and the mean model-based standard error, and nominal-95% CI coverage. Strict RM-ANOVA is excluded from these coefficient-based measures because its estimand is on the dichotomized scale.

The full Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures are pre-registered at analysis/scripts/test-procedure-design-sensitivity/00-ademp-pr which fixes the test-procedure times design-grid factorial across three studies (Study 1: test-procedure comparison at fixed design; Study 2: cycle-by-period grid at fixed test, 240 cells; Study 3: joint design-by-test heatmap, 240 cells) before the production runs commit. The present paper executes a revised Study 1; the executed design departs from the pre-registered Study 1 on several factor levels, and we document every departure in Table 1 below in the interest of full ADEMP transparency.

Table 1: Pre-registered versus executed Study 1. Departures are recorded for transparency; the executed superset of sample sizes (which restores the pre-registered N levels and adds $N = 25$) and the added continuous-biomarker contrast arm respond to referee comments on the first internal draft.

Factor	Pre-registered	Executed (this paper)
Test procedures	3 (RM-ANOVA, LME, GEE-MD)	5 (adds naive GEE and the continuous-biomarker contrast)
Biomarker effect c_{bm}	$\{0, 0.3, 0.6\}$	$\{0, 0.45\}$ (the project-wide reference effect size)
Sample size N	$\{35, 70, 100\}$	$\{25, 35, 70, 100\}$ (superset; adds $N = 25$)
Replicates per cell	1000 (5000 for Type I cells)	1000 power, 2000 Type I

Factor	Pre-registered	Executed (this paper)
Carryover half-life	$t_{1/2} = 3$ days	$t_{1/2} = 0$ (no carryover at the fixed reference)

In the scope of the present paper (test-procedure axis only at the reference design), the simulation runs on the `pmsimstats` package data-generating process under Architecture A (mean moderation), exercised through `generateData()` and the standard `lme_analysis()` fit, rather than on the reduced `implementations/simple` substrate. The factorial comprises $n_{\text{reps}} = 1000$ per power cell and 2000 per Type I cell, across the 40 cells defined by procedure $\in \{\text{strict RM-ANOVA, continuous-biomarker contrast, linear-mixed with corCAR1, GEE with naive sandwich, GEE with Mancl-DeRouen small-sample correction}\} \times c_{bm} \in \{0, 0.45\} \times N \in \{25, 35, 70, 100\}$. Execution uses `parallel::mclapply` with `RNGkind("L'Ecuyer-CMRG")` and `mc.set.seed = TRUE`, so each replicate draws from an independent, reproducible random-number stream. The Mancl-DeRouen correction was implemented inline (no `geesmv`-style external dependency in the production path) by inflating each cluster's residual via $(I - H_{ii})^{-1} r_i$ in the sandwich meat. It produces identical point estimates and larger standard errors than the naive sandwich, and its SE for the interaction coefficient agrees with `geesmv::GEE.var.md` to within approximately 2.5% (mean inline/`geesmv` ratio 1.02, standard deviation 0.02, over 25 probe replicates at $N = 35$, $c_{bm} = 0.45$), so the correction magnitude is externally validated rather than merely plausible.

6 Results

The simulation program was executed across the 40 cells of the five-procedure-by-effect-by- N factorial in 587 s using 7-core `parallel::mclapply` with `RNGkind("L'Ecuyer-CMRG")` per-stream seeding. Convergence was 100% across all 60{,}000 fits. Per-replicate output is at `analysis/data/quick-sim/08-test-replicates.rds`; the Morris ADEMP cell-level summary is at `analysis/data/quick-sim/08-test-summary.txt`. With 1000 replicates per power cell the MCSE on a power estimate near 0.5 is approximately 0.016, and with 2000 replicates per Type I cell the MCSE on a rate near 0.05 is approximately 0.005. We begin the results with the test-procedure comparison at the fixed reference design.

6.1 Test-procedure comparison at fixed design

We note at the outset that strict RM-ANOVA targets a different estimand from the other four procedures. It operates on a median-dichotomized biomarker, pre-averaged to one value per phase, and so estimates a dichotomized-stratum interaction; the continuous-biomarker contrast, the linear-mixed model, and both GEE variants estimate the interaction coefficient $\beta_{bm:D}$ on the continuous symptom scale. An earlier version of this comparison contrasted strict RM-ANOVA directly with the mixed model and attributed the resulting power gap to the test

procedure; that attribution conflated two distinct costs. We therefore introduce the continuous-biomarker contrast of §4, which differs from strict RM-ANOVA only in retaining the continuous biomarker (it shares the pre-averaging and the absence of any AR(1) machinery) and differs from the linear-mixed model only in discarding the within-phase longitudinal structure. The RM-ANOVA $\rightarrow$ contrast step therefore measures the cost of dichotomization in isolation, and the contrast $\rightarrow$ linear-mixed step measures the cost of the test procedure proper. Reading the two steps separately is the honest decomposition the earlier direct comparison lacked.

Type I error at the null cell ($c_{bm} = 0$), percent $\pm$ MCSE, is shown in Table 2; power at $c_{bm} = 0.45$ in Table 3.

Table 2: Type I error rate (percent $\pm$ MCSE) under $H_0 : \beta_{bm:D} = 0$, 2000 replicates per cell. Nominal level 5%.

Procedure	$N = 25$	$N = 35$	$N = 70$	$N = 100$
Strict RM-ANOVA (dich.)	4.55 ± 0.47	4.15 ± 0.45	5.95 ± 0.53	5.20 ± 0.50
Contrast (continuous bm)	5.25 ± 0.50	4.50 ± 0.46	6.45 ± 0.55	4.60 ± 0.47
Linear-mixed (corCAR1)	2.75 ± 0.37	2.05 ± 0.32	3.35 ± 0.40	2.90 ± 0.38
GEE (naive sandwich)	9.70 ± 0.66	8.55 ± 0.63	7.90 ± 0.60	7.40 ± 0.59
GEE (Mancl-DeRouen)	6.35 ± 0.55	5.30 ± 0.50	6.25 ± 0.54	5.90 ± 0.53

Table 3: Power (percent $\pm$ MCSE) at $c_{bm} = 0.45$, 1000 replicates per cell.

Procedure	$N = 25$	$N = 35$	$N = 70$	$N = 100$
Strict RM-ANOVA (dich.)	35.8 ± 1.52	46.8 ± 1.58	75.3 ± 1.36	91.3 ± 0.89
Contrast (continuous bm)	53.8 ± 1.58	67.7 ± 1.48	91.5 ± 0.88	98.8 ± 0.34
Linear-mixed (corCAR1)	56.5 ± 1.57	71.2 ± 1.43	94.5 ± 0.72	99.2 ± 0.28
GEE (naive sandwich)	67.2 ± 1.48	75.9 ± 1.35	94.5 ± 0.72	99.2 ± 0.28
GEE (Mancl-DeRouen)	57.3 ± 1.56	69.1 ± 1.46	92.7 ± 0.82	99.1 ± 0.30

Four substantive findings emerge from the tables. First, and most consequentially for how the comparison should be read, the power gap between strict RM-ANOVA and the linear-mixed model is predominantly a cost of dichotomizing the

433 biomarker rather than a property of the test procedure. At $N = 25$ the strict RM-
 434 ANOVA power is 0.358; the continuous-biomarker contrast, which differs only in
 435 retaining the continuous biomarker, reaches 0.538, so dichotomization alone ac-
 436 counts for 0.180 of lost power. The further step to the full linear-mixed model
 437 raises power only to 0.565, an increment of 0.027 attributable to the test proce-
 438 dure proper. The decomposition is stable across the grid: the dichotomization
 439 component is 0.209 at $N = 35$ ($0.468 \rightarrow 0.677$), 0.162 at $N = 70$ ($0.753 \rightarrow 0.915$),
 440 and 0.075 at $N = 100$ ($0.913 \rightarrow 0.988$), while the contrast-to-linear-mixed incre-
 441 ment never exceeds 0.04. The earlier framing, which compared strict RM-ANOVA
 442 against the mixed model directly and read the difference as a test-procedure ef-
 443 fect, therefore overstated the role of the test procedure by roughly a factor of five;
 444 the honest statement is that a classical, sphericity-free, AR(1)-free within-subject
 445 contrast on the continuous biomarker captures most of the available power, and
 446 the mixed model adds a small further increment.

447 Second, GEE with the naive sandwich is anti-conservative across the entire sample-
 448 size range examined. Its null rejection rate is 0.097 at $N = 25$ and declines only
 449 slowly with N to 0.086, 0.079, and 0.074 at $N = 35, 70, 100$ – still roughly $1.5\times$
 450 nominal at $N = 100$. The inflation is therefore not a small-cluster curiosity that
 451 disappears at trial-relevant sizes; it persists. The Mancl-DeRouen 2001 small-
 452 sample sandwich correction brings the rate within or close to MCSE of nominal
 453 at every N (0.064, 0.053, 0.063, 0.059), at a power cost of about ten percentage
 454 points at $N = 25$ ($0.672 \rightarrow 0.573$) and seven at $N = 35$ ($0.759 \rightarrow 0.691$). The
 455 corrected procedure’s apparent power advantage over the linear-mixed model at
 456 small N (Table 3) is the residue of its remaining slight over-rejection, not a genuine
 457 efficiency gain.

458 Third, the secondary estimands (Table 4) locate the procedures’ behavior in the
 459 standard-error calibration that the rejection rates only summarize. At $c_{bm} =$
 460 0.45, the continuous-biomarker contrast and the Mancl-DeRouen GEE are well-
 461 calibrated: their nominal-95% CI coverage is 93–95% and their mean model-based
 462 SE is within a few percent of the empirical SE. Naive GEE under-covers at small
 463 N (90.0% at $N = 25$, 91.5% at $N = 35$) because its model-based SE is 9 to 12%
 464 smaller than the empirical SE, the calibration counterpart of its Type I inflation.
 465 The linear-mixed model over-covers (97.0 to 97.7%) because its model-based SE
 466 exceeds the empirical SE by 11 to 16%; this is the same conservatism visible in
 467 its sub-nominal Type I error.

Table 4: Secondary estimands at $c_{bm} = 0.45$ for the four coefficient-bearing arms (the interaction estimand is $\beta_{bm:D} = -2.21$, the Monte-Carlo reference value). Coverage is of the nominal-95% Wald interval; SE ratio is mean model-based SE divided by empirical SE.

Arm	Coverage $N = 25$	Coverage $N = 100$	SE ratio $N = 25$	SE ratio $N = 100$	Max rel. bias
Contrast (continuous bm)	94.9	94.8	0.99	0.98	3.0%
Linear-mixed (corCAR1)	97.7	97.0	1.16	1.11	2.6%
GEE (naive sandwich)	90.0	93.3	0.91	0.98	3.8%
GEE (Mancl- DeRouen)	94.3	94.0	1.07	1.02	3.8%

Fourth, the comparison is not contaminated by point-estimate bias. The relative bias of $\hat{\beta}_{bm:D}$ is below 4% of the estimand $|\beta_{bm:D}| = 2.21$ for the contrast, the linear-mixed model, and both GEE variants in every cell, so the differences in power and coverage reflect variance estimation and information use rather than systematic mis-estimation. The linear-mixed procedure’s Type I error sits below nominal throughout (0.021 to 0.034), the conservatism already visible in its over-coverage and its model/empirical SE ratio above one; it is consistent with **corCAR1** degrees-of-freedom accounting under the short ($T = 8$) serial series. A Kenward-Roger²⁸ or Satterthwaite²⁹ small-sample degrees-of-freedom correction would address the conservatism and recover the small power increment it forgoes; this extension is recorded as a follow-up item.

7 Design-and-test recommendation

The recommendation table below maps test procedure choice to typical chronic-condition N-of-1 trial scenarios at the prazosin/PTSD reference design (Hendrickson hybrid OL+BDC, biomarker-treatment interaction estimand). The cycle count and period length are held at the reference values defined in §3 because the joint optimization over those axes is deferred to the follow-up paper. We present the recommendation as a table of preferred (procedure, scenario) combinations rather than as a single ‘best’ procedure, because the preferred choice depends on features of the application that the analyst must determine. The overriding recommendation, however, is design-agnostic: retain the biomarker as a continuous predictor, since dichotomization forfeits the large majority of the available power (§5, Table 3).

Scenario	Preferred procedure	Rationale
Default (continuous biomarker, any N)	linear-mixed (corCAR1)	Highest power within nominal Type I; corCAR1 matches AR(1) physiology; well-behaved across N
Lightweight / transparent classical analysis	continuous-biomarker contrast	Within 0.03 of the mixed model on power, nominal Type I, no covariance modeling required
$N \leq 35$, concern about LME conservatism	linear-mixed (corCAR1) with Kenward-Roger	LME sub-nominal and over-covers; KR addresses this without inflating Type I
Robust-SE-required setting (e.g. unmodeled clustering)	GEE Mancl-DeRouen	Restores nominal Type I and 93–95% coverage; modest power cost
Reviewer-mandated ‘classical’ analysis	strict RM-ANOVA	Dichotomized sensitivity comparator; report alongside a continuous-biomarker analysis with the power-loss decomposition
Naive GEE (no small-sample correction)	not recommended	Type I error $1.5\text{--}1.9\times$ nominal across all N ; under-covers

491 The dominant message is that the biomarker should be analyzed as a continuous
 492 predictor: dichotomizing it for a strict RM-ANOVA forfeits the large majority of
 493 the power gap that the present study isolates, while modeling choices among the
 494 continuous-biomarker procedures move power only at the margin. Within that
 495 class, the linear-mixed analysis with continuous-time AR(1) residual correlation
 496 is the preferred default for the biomarker-treatment interaction in aggregated N-
 497 of-1 trials at the parameter scale evaluated here, with the dichotomization-free
 498 within-subject contrast a close and transparent alternative that requires no covari-
 499 ance modeling. Strict RM-ANOVA remains useful as a dichotomized sensitivity
 500 comparator, as the §2 closed-form derivation makes explicit, but it should be
 501 reported alongside a continuous-biomarker analysis and the power-loss decompo-
 502 sition rather than on its own. Naive GEE should not be used at the sample
 503 sizes examined without a small-sample correction; the Mancl-DeRouen correction
 504 restores nominal Type I error and 93–95% coverage at a moderate power cost.

8 Discussion

8.1 What the Monte Carlo study settles

We begin the discussion by summarizing what the Monte Carlo study settles at the resolution it was run. The run on the 40-cell five-procedure factorial (procedure $\in \{\text{strict RM-ANOVA, continuous-biomarker contrast, linear-mixed with corCAR1, GEE with naive sandwich, GEE with Mancl-DeRouen small-sample correction}\} \times c_{bm} \in \{0, 0.45\} \times N \in \{25, 35, 70, 100\}$, with 1000 replicates per power cell and 2000 per Type I cell) was executed in 587 s with 100% convergence across all 60{,}000 fits. Per-replicate output is at `analysis/data/quick-sim/08-test-replicates.rds`; Morris ADEMP summary at `08-test-summary.txt`. With 1000 power replicates the MCSE on a power estimate near 0.5 is approximately 0.016 and with 2000 Type I replicates the MCSE on a rate near 0.05 is approximately 0.005. The Mancl-DeRouen 2001²⁴ correction was implemented inline by inflating each cluster’s residual via $(I - H_{ii})^{-1}r_i$ in the sandwich meat; it produces identical point estimates and larger standard errors than the naive sandwich, and its SE for the interaction coefficient agrees with `geesmv::GEE.var.md` to within approximately 2.5% (mean inline/`geesmv` ratio 1.02 over 25 probe replicates), so the correction magnitude is externally validated.

9 Conclusions

In conclusion, three points are to be emphasized. First, the analyst’s most consequential choice for the biomarker-treatment interaction in aggregated N-of-1 trials is to keep the biomarker continuous rather than to dichotomize it for a classical RM-ANOVA. The continuous-biomarker contrast recovers about four-fifths of the apparent RM-ANOVA-to-mixed-model power gap (dichotomization costs 0.08 to 0.21 of power across the sample sizes evaluated, while the test procedure proper adds at most 0.04); the linear-mixed analysis with continuous-time AR(1) residual correlation has the highest power within nominal Type I among the continuous-biomarker procedures, with the dichotomization-free within-subject contrast a close alternative. Naive GEE with the standard sandwich estimator inflates Type I error to between $1.5\times$ and $1.9\times$ nominal across the whole $N \in \{25, 35, 70, 100\}$ range, disqualifying its rejection rate as a power summary; the Mancl-DeRouen 2001 small-sample sandwich correction restores nominal Type I control and 93–95% coverage at a moderate (about 10-percentage-point) power cost, making it the operational GEE comparator rather than the naive sandwich.

Second, the closed-form strict-RM-ANOVA test of the biomarker-treatment interaction (§2) reduces to a two-sample t -test on the within-subject treatment differences under sphericity and balance. The reduction makes the strict-RM-ANOVA result analytically transparent and connects the simulation findings to the classical split-plot literature. It also pins down what does and does not limit the test here. The strict RM-ANOVA arm we evaluate pre-averages each phase to a single value, so its within-subject covariance is 2×2 and sphericity holds by construc-

tion; consistent with this, its Type I error is at nominal (Table 2), and its power deficit is wholly attributable to the median dichotomization of the biomarker, as the continuous-biomarker contrast demonstrates. The sphericity penalty that motivates much of §2 applies instead to the alternative classical analysis that treats the full $T = 8$ series as repeated within-subject levels: there the AR(1) residual correlation violates sphericity and a Greenhouse-Geisser correction recovers nominal Type I only at a degrees-of-freedom cost, which the linear-mixed analysis with `corCAR1` avoids by modeling the AR(1) structure directly. Both considerations point the same way – away from a dichotomized classical analysis – but through different mechanisms, and we are careful to keep them distinct.

Third, the cycle-by-period design grid that constitutes the second axis of the joint design-by-test sensitivity question framed in §1 is deferred to a follow-up paper. The infrastructure for the sweep is in place at `analysis/scripts/test-procedure-design-sensitivity/`, and the design specification is committed at `docs/27-cycle-period-design-sweep-plan.md`. The follow-up paper will deliver the joint (cycle count, period length, test procedure) recommendation table that the present paper’s §6 narrows to the test-procedure-only axis.

10 Conflict of interest

The authors declare no conflicts of interest.

11 Funding

This work received no specific funding.

12 Data availability statement

The data and code that support the findings of this study are openly available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths relevant to this manuscript are listed in the Reproducibility section below.

13 Reproducibility

This manuscript’s simulation was executed with R 4.6.x; the project also ships a `zzcollab` Docker container (image hash recorded in `Dockerfile`; package set captured in `renv.lock`) for container-based reproduction. Principal packages used by the Study 1 pipeline are `nlme` (continuous-time linear-mixed analysis with `corCAR1` residual correlation), `geepack` (GEE fitting) plus an inline Mancl-DeRouen 2001 small-sample sandwich correction, `geesmv` (independent benchmark of that correction), `data.table`, and `parallel` (`mclapply` for the

parameter-grid sweep), with `rmarkdown` plus `bookdown` for the manuscript build.
 The Study 1 simulation runs on the `pmsimstats` package data-generating process
 under Architecture A via `generateData()` and `lme_analysis()`, not on the
 reduced implementations/simple substrate.

Random-number generator. The driver sets `RNGkind("L'Ecuyer-CMRG")`
 and `set.seed(20260507)`, then runs the cell-by-replicate jobs through
`parallel::mclapply` with `mc.set.seed = TRUE`, so each replicate draws
 from an independent, reproducible random-number stream.

Data and code availability. Per-replicate simulation outputs are check-
 pointed at `analysis/data/quick-sim/08-test-replicates.rds`; the extended
 Morris ADEMP cell-level summary (rejection rate, MCSE, convergence, bias,
 empirical and model SE, coverage) at the companion `08-test-summary.txt`,
 with the Monte-Carlo reference estimand at `08-test-betatruetext` and the
 Manc1-DeRouen benchmark at `08-test-manc1-benchmark.txt`. The driver is
`analysis/scripts/test-procedure-design-sensitivity/01-study1-test-procedure.R`
 (superseding the earlier `analysis/scripts/quick-sim/08-test-procedure-prototype.R`).
 The pre-registered ADEMP plan, and the deviations recorded in Table 1, are at
`analysis/scripts/test-procedure-design-sensitivity/00-ademp-pre-registration.md`.
 The manuscript source, paper-specific bibliography (`references.bib`), and SIM
 CSL file are at `analysis/report/08-test-procedure-design-sensitivity/`.

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